

Counterexamples to Detlefs' Two Fibonacci–Fermat Characterisations of the Primes, OEIS A000040

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Abstract

In May 2014, G. Detlefs contributed to OEIS A000040 two conjectural characterisations of the primes, each combining a Fibonacci condition $F(n) \equiv \pm 1 \pmod{n}$ with a base-2 Fermat condition. Both are false. The smallest counterexample to each is $n = 219781 = 271 \cdot 811$, and there are nineteen counterexamples below $3 \cdot 10^6$. The failure is structural, not accidental: by Catalan's identity the Fibonacci condition is, up to the choice of a square root of 1, the Fibonacci-pseudoprime condition $F(n - (\frac{5}{n})) \equiv 0$ in the case $(\frac{5}{n}) = +1$, and Pomerance, Selfridge and Wagstaff restrict their well-known \$620 problem to the case $(\frac{5}{n}) = -1$ precisely because the other case admits easy counterexamples. Detlefs imposes no condition modulo 5 and so lands in the easy case. The counterexamples themselves are not new — the first four appear in the 1980 tables of Pomerance, Selfridge and Wagstaff — but they appear not to have been connected to these two comments, which still stand on the entry.

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1 The two conjectures

Let F denote the Fibonacci sequence, $F(0) = 0$, $F(1) = 1$. On 25 and 28 May 2014, Gary Detlefs contributed to OEIS A000040, the sequence of primes, the following two comments [2]:

(C4) *Conjecture: Sequence = {5 and $n \neq 5$ | (Fibonacci(n) mod n = 1 or Fibonacci(n) mod n = $n - 1$) and 2^{n-1} mod n = 1}.*

(C5) *Conjecture: Sequence = {5 and $n \neq 5$ | (Fibonacci(n) mod n = 1 or Fibonacci(n) mod n = $n - 1$) and 2^{3n} mod $3n$ = 8}.*

Both assert that a certain set of integers is exactly the set of primes. The prime 5 is excluded and re-added by hand because $F(5) = 5 \equiv 0 \pmod{5}$. Both comments were present, unrebutted, on the live entry on 22 August 2026; the page carried the timestamp 25 July 2026.

The forward direction is unproblematic. If p is prime then $2^{p-1} \equiv 1 \pmod{p}$ for p odd, and $F(p) \equiv (\frac{5}{p}) \pmod{p}$, which is ± 1 for $p \neq 5$. What fails is the converse.

Theorem 1. *Let $n = 219781 = 271 \cdot 811$. Then n is composite and*

$$2^{n-1} \equiv 1 \pmod{n}, \quad 2^{3n} \equiv 8 \pmod{3n}, \quad F(n) \equiv 1 \pmod{n}.$$

Hence n lies in the set described by (C4) and in the set described by (C5), but not in A000040. Both conjectures are therefore false. Moreover n is the smallest such integer.

Proof. The factorisation $219781 = 271 \cdot 811$ is immediate, and both factors are prime. The three congruences are single modular exponentiations; they are carried out in the accompanying script, which computes $F(n) \bmod n$ by the fast-doubling identities

$$F(2m) = F(m)(2F(m+1) - F(m)), \quad F(2m+1) = F(m)^2 + F(m+1)^2.$$

Minimality is by exhaustive search over all odd composite $n < 3 \cdot 10^6$. □

The nineteen counterexamples below $3 \cdot 10^6$ are

219781, 252601, 399001, 512461, 722261, 741751, 852841, 1024651,
1193221, 1533601, 1690501, 1735841, 1857241, 1909001, 2100901,
2165801, 2603381, 2704801, 2757241.

Each satisfies both (C4) and (C5), each has $F(n) \equiv +1 \pmod{n}$, and each satisfies $n \equiv 1 \pmod{5}$. The first four appear in the tables of Pomerance, Selfridge and Wagstaff [1] as the smallest integers that are simultaneously base-2 Fermat pseudoprimes and Lucas pseudoprimes for the Fibonacci parameters $(P, Q) = (1, -1)$. The numbers are thus long known; what appears not to have been noticed is that they refute these two comments.

2 Why the conjectures fail

The counterexamples are not sporadic. Two observations locate the gap exactly.

Lemma 2. *If n is odd and $F(n-1) \equiv 0 \pmod{n}$, then $F(n)^2 \equiv 1 \pmod{n}$.*

Proof. Catalan's identity in the form $F(n-1)F(n+1) - F(n)^2 = (-1)^n$ gives, for odd n , $F(n-1)F(n+1) - F(n)^2 = -1$. Reducing modulo n and using $F(n-1) \equiv 0$ leaves $-F(n)^2 \equiv -1$, that is $F(n)^2 \equiv 1$. □

So for an odd n satisfying the Fibonacci-pseudoprime congruence $F(n - (\frac{5}{n})) \equiv 0 \pmod{n}$ with $(\frac{5}{n}) = +1$, the quantity $F(n)$ is automatically a square root of 1 modulo n ; that it is the particular square root $+1$ rather than one of the other roots available for composite n is a further condition, but an easily met one, and it is met by every counterexample above. Detlefs' Fibonacci condition is therefore not a new test: it is the classical Fibonacci-pseudoprime test in the split case.

Remark 3. Pomerance, Selfridge and Wagstaff [1] offer \$620 for a composite $n \equiv \pm 2 \pmod{5}$ that is simultaneously a base-2 pseudoprime and a Fibonacci pseudoprime, or for a proof that none exists. The restriction $n \equiv \pm 2 \pmod{5}$, equivalently $\left(\frac{5}{n}\right) = -1$, is not decorative: it is imposed precisely because in the complementary case $\left(\frac{5}{n}\right) = +1$ such composites exist and are small. Conjectures (C4) and (C5) impose no condition modulo 5 and hence admit exactly those composites. Every counterexample listed above has $n \equiv 1 \pmod{5}$, as the analysis predicts.

Adding the hypothesis $n \equiv \pm 2 \pmod{5}$ would turn (C4) into a variant of the PSW problem. That variant is not settled here and is not settled by anyone; the present note refutes the conjectures only as stated.

Remark 4. The two conjectures differ only in their Fermat clause, and the difference is cosmetic. For odd n coprime to 3, the condition $2^{3n} \equiv 8 \pmod{3n}$ is equivalent to $2^{3n-3} \equiv 1 \pmod{n}$ together with the automatic congruence modulo 3; in particular every base-2 Fermat pseudoprime in the list satisfies it, as verified directly. A single counterexample therefore kills both.

3 Computational verification

A short script (Python 3, standard library plus `sympy`) performs the following; all checks pass.

1. Exhaustive search over odd composite $n < 3 \cdot 10^6$ returns exactly the nineteen values listed above satisfying $2^{n-1} \equiv 1 \pmod{n}$ and $F(n) \equiv \pm 1 \pmod{n}$. In particular 219781 is the smallest.
2. For each of the nineteen, the script records the factorisation and the values of $2^{n-1} \pmod{n}$, $2^{3n} \pmod{3n}$, $F(n-1) \pmod{n}$, $F(n) \pmod{n}$ and $n \pmod{5}$. In every case these are 1, 8, 0, 1 and 1 respectively.
3. Control: the three congruences of Theorem 1 hold for the primes 7, 11, 13, 101, 10007, 104729, with $F(p) \pmod{p}$ equal to 6, 1, 12, 1, 10006, 1 — that is ± 1 in each case, as $\left(\frac{5}{p}\right)$ predicts.

Every numeral in this note is an output of that script.

4 Concluding remarks

The disproof is a single modular exponentiation and the numbers involved have been in print since 1980; we claim no difficulty for any of it. It is recorded because the two comments have stood on A000040 for eleven years, because the set of counterexamples is not thin, and because the reason for the failure — the missing congruence condition modulo 5 — is worth stating alongside them, so that the comments are not merely deleted but corrected. The natural destination is comments on the entry rather than a journal.

We have found no earlier notice on the entry or in the literature that these two comments fail, but the observation is short enough that prior knowledge would not be surprising, and we would welcome notice of any earlier appearance.

References

- [1] C. Pomerance, J. L. Selfridge and S. S. Wagstaff, Jr., *The pseudoprimes to $25 \cdot 10^9$* , Mathematics of Computation **35** (1980), no. 151, 1003–1026; and the companion table of Lucas pseudoprimes, Math. Comp. **35** (1980), no. 152, 1391–1417.
- [2] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A000040 (the primes); comments of G. Detlefs, 25 and 28 May 2014.